AMITY UNIVERSITY
PATNA

Course Curriculum

Course Title: Time Series Analysis
Course Code: STAT311
Credit Units: 05
Course Level: UG
Syllabus Version – I

L	T	P/S	SW	No. of PSDA	Total Credit Unit
3	0	4	0	0	5

Course Description

This course covers the methodology and applications of time series analysis and forecasting, focusing on issues and problems predicting business and economic data. The course is intended to serve as a guide to the principles, assumptions, strengths, limitations, and application of time series models and forecasting methods. This course introduces concepts essential to understanding the rationale of time series analysis. It reviews basic statistical principles and techniques that form the foundation for learning about time series methods. It covers most of the core techniques currently used in time series analysis. Data evaluation, identification of suitable models, estimation methods, and assessment of the models are explained with examples. Many examples of the application of time series analysis and forecasting to the problems in various business domains, including marketing, retail sales, human resource management, operations and supply chain management, finance, and general management**.**

Course Objectives:

CO1: Understand foundational concepts of time series data, including stationarity, autocorrelation, seasonality, and trend components.
CO2: Apply statistical techniques for time series decomposition and smoothing methods such as moving averages and exponential smoothing.
CO3: Analyze Use software tools (such as R, Python, or MATLAB) to analyze and visualize time series data.
CO4: Evaluate Perform diagnostic checks to validate the adequacy and accuracy of time series models.

Prerequisites:

To perform effective **time series analysis**, several **prerequisites**—in terms of knowledge, data preparation, and tools—should be in place.

Course Contents/Syllabus:

	Weightage (%)	Lecture Hour
Module I:	**25**	**11**
Introduction to R and Python infrastructure for time series analysis.		
Module II:	**25**	**11**
. **Essential characteristics of the time-series data**- Time Series and Their Features; Basic Descriptive Techniques.Trends and Time Series Decomposition.		
Module III:	**25**	11
. **Forecasting Time Series and Smoothing-** Smoothing Techniques, Introduction to Linear Time Series Models. ARIMA Modeling.		
Module IV:	**25**	**12**
. **Models for Time series and forecasting-** Volatility Models. State – Space Representation of the time series.Time-Series forecasting and performance Evaluation.		

Course Learning Outcomes(CLOs):

CLO1: Understand the fundamental concepts of time series data**,** including stationarity, autocorrelation, and seasonality.

CLO2: Apply appropriate statistical methods for analyzing time series data, such as decomposition, smoothing techniques (e.g., moving average, exponential smoothing), and ARIMA modeling.

CLO3: Identify and estimate time series models, including AR, MA, ARMA, ARIMA, and SARIMA.

CLO4: Diagnose model adequacy using residual analysis and goodness-of-fit tests.

CLO5: Forecast future values using fitted time series models and evaluate forecast accuracy.

Assessment/ Examination Scheme:

Theory L/T (%)	**Lab/Practical/Studio (%)**	**TOTAL**
60	**40**	**100**

Theory Assessment (L&T):

Continuous Assessment/Internal Assessment (40 %)					End Term Examination (60%)
Components (Drop down)	Viva	Class Test	Home Assignment	Attendance	End Term
Weightage (%)	5	20	10	5	60

Lab/ Practical/ Studio Assessment:

Continuous Assessment/Internal Assessment (40 %)						End Term Examination (60%)	
Components (Drop down)	Class Test (Practical based)	Lab Record	Continuous Performance	Internal Viva	Attendance	Practical Evaluation	Viva
Weightage (%)	10	10	10	5	5	40	20

Time Series Analysis: Lab Problems [60 Hr.]

S.No.	**Experiment**	**Practical Hours**
1	A company is experiencing a seasonal pattern in their sales data. Develop a time series model (e.g., ARIMA) to forecast sales for the next year, considering seasonality.	6
2	You have a time series dataset of daily stock prices. Analyze the data to identify trends, seasonality, and cyclical patterns. Use appropriate techniques (e.g., moving averages, decomposition) to isolate and understand these components	6
3	A city wants to predict the number of tourists visiting each month. Analyze historical tourism data to identify patterns and forecast future tourist arrivals	6
4	A factory wants to optimize production based on historical electricity consumption data. Analyze the data to identify patterns and predict future energy consumption	6
5	Explain how time series analysis can be used in weather forecasting	6
6	Describe how time series analysis can be used to analyze financial data.	6
7	How would you use your trained ARIMA model (or other model) to forecast future values in your time series?	6
8	How would you generate confidence intervals for your forecasts?	6

9	How would you evaluate the accuracy of your forecasts using appropriate evaluation metrics?	6
10	You have a dataset of daily stock prices. How would you load this data into R or Python, and what preliminary exploration (plotting, descriptive statistics) would you perform to understand its characteristics?	6

Mapping Continuous Evaluation components/PSDA with CLOs					
SN	**Bloom's Level**	**Course Learning Outcomes**	**Class Test**	**Home Assignment**	**Viva**
1	**Understand**	Understand foundational concepts of time series data, including stationarity, autocorrelation, seasonality, and trend components.	✓	✓	✓
2	**Apply**	Apply statistical techniques for time series decomposition and smoothing methods such as moving averages and exponential smoothing.	✓	✓	✓
3	**Analyze**	Use software tools (such as R, Python, or MATLAB) to analyze and visualize time series data.	✓	✓	✓
4	**Evaluate**	Perform diagnostic checks **t**o validate the adequacy and accuracy of time series models.	✓	✓	✓

Text Books:

1. Box, G.E.P., Jenkins, G.M. and Reinsel, G.C. (1994) Time Series Analysis, Forecasting and Control, Englewood Cliffs, NJ: Prentice-Hall.
2. Chris Chatfield, and Haipeng Xing (2019) The Analysis of Time Series: An Introduction with R. CRC Press, London.
3. Galit Shmueli and Kenneth C. Lichtendahl Jr (2016). Practical Time Series Forecasting with R: A hands-on Guide, Axelrod Schnall Publishers.

Reference Book

1. "Introductory Time Series with R (Use R!)" **Authors:** Paul S.P. Cowpertwait and Andrew V. Metcalfe
2. "The Analysis of Time Series: An Introduction" **Author:** Chris Chatfield
3. "Time Series Analysis" **Author**: James Douglas Hamilton
4. "Forecasting: Principles and Practice" **Authors:** Rob J. Hyndman and George Athanasopoulos